

SpectreFLY Design Document

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Abstract—This document aims to teach the reader how to create the SpectreFLY FireWatch Drone.
Index Terms—none

Executive Summary

SpectreFLY is a rapidly deployable quadcopter platform designed as a first-response environmental monitoring tool for high wildfire-risk areas with little or no sensor infrastructure. The system integrates temperature, humidity, and smoke sensing with live data transmission to a ground station, enabling early awareness and alerts when fire-risk is detected. Its mechanical and electrical architecture allows it to adapt to different terrains and sensing requirements.

The design section of the document specifies system requirements for payload capacity, flight time, communication range, and sensor capability. It targets operation in terrain suited poorly for manual monitoring. The architecture defines the flight controller, power distribution, propulsion stack, communication link, and sensor bus. The mechanical design focuses on a 3D-printed frame with mounting points for motors, batteries, and payloads.

Power and propulsion modules are chosen to maximize flight time, while keeping a stable rate of power consumption. Sensor and payload modules include environmental sensors that measure temperature, humidity, and smoke concentration to support fire monitoring. The electronics and control subsection describes how the flight controller interfaces with ESC, receivers, and sensors over buses.

Safety and reliability are assured through design features and planned validation. Propeller guards and keyed connectors prevent user error and physical hazards. The document outlines bench testing for individual modules (power, propulsion, sensors, communications) and flight testing for stable hover, controlled movement, data link, and detection of smoke/temperature. These elements aim to demonstrate that SpectreFLY can serve as a practical platform for wildfire monitoring and emergency response research.

1 Introduction

This document contains all the information needed to design and use the SpectreFLY Modular Drone. [more later]

1.1 Need Statement

A rapidly deployable drone that can be launched immediately into areas with zero sensor infrastructure, collecting real-time temperature, humidity, and smoke data during the high-risk window before a fire occurs.

1.2 Goal Statement

Prove that a drone can function as a first response, environmental monitoring tool by integrating a smoke and temperature sensor, transmitting live readings to a ground station, and triggering alerts when fire risk thresholds are crossed.

1.3 Personas

1.4 Existing Research

Currently little to no versatile, quickly deployable solution to monitoring wildfires in areas without extensive fire monitoring.

1.5 Sustainability Statement

2.8 Testing and Validation

[bench tests for each module.]
[flight tests and criteria.]

2 Design

2.1 System Requirements

[functional requirements.]
[payload, flight time, range, sensor capabilities.]

2.2 System Architecture

[overall block diagram.]
[flight controller, power, propulsion, communication, sensor bus.]

2.3 Mechanical Design

[frame layout and mounting points for motor, battery, ...]

2.4 Power and Propulsion Modules

[information about motors/propellers.]

2.5 Sensor and Payload Modules

[information about supported sensors (LiDAR, camera, ...), maximum excess payload, etc.]

2.6 Electronics and Control

[how to control the drone, what is needed for control, etc.]

2.7 Safety and Reliability

[testing logs and proof that it can take damage and handle failures.]

[physical safety information: prop guards, connector keying to avoid misplugging.]

3 Parts List

Part	Set	Supplier	Quantity
CarbonX PLA+CF Filament	A	Creality	1
Battery: LiPo 850 mAh 3S 25/30C	A	HobbyKing	1
Motor: SunnySky R1106 FPV Brushless Motors	A	Druav	4
Propeller: Gemfan Hurricane 2009-3 Tri-Blade (Set of 4)	A	Pyrodrone	1
Battery: 3S 1000 mAh	B	HobbyKing	1
Motor: RCINPOWER GTS 1204 5000KV	B	Pyrodrone	4
Propeller: HQProp T3.5x2x3	B	GetFPV	1
Caddx Ant Lite Analog Camera (FPV Cycle Edition)	A / B	CaddxFPV	1
TFMini-S Micro LiDAR Module	A / B	SparkFun	1
HAKRC 20A 4-in-1 3-6S BLHeli_32 5V 3A BEC DShot1200 Mini ESC	A / B	Alibaba	1
Perforated Circuit Board	A / B	Adafruit	2
Female Header Pins 2x6	A / B	AliExpress	4
Female Header Pins 2x8	A / B	AliExpress	2
Female Header Pins 2x10	A / B	Mikroe	2

TABLE I: Itemized Budget